

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Examination Summer 2023

Course: Second Year B. Tech. V (Sem – IV)

Branch: Civil Engineering

Subject Name: Hydraulics - II

Subject Code: BTCVC405

Max Marks: 60

Duration: 3 Hrs.

Date: 22/07/2023

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|----|--|----------|----|
| A) | Explain classification of flow in open channel and Types of channels. | Remember | 06 |
| B) | Write a note on velocity distribution in open channel and explain pitot tube. | CO 1 | 06 |
| C) | Water flows over a rectangular weir 1.5 m wide at a depth of 160 mm and afterwards passes through a triangular right-angled weir. Taking C_d for the rectangular and triangular weir as 0.62 and 0.59 respectively. Find the depth over the triangular weir. | CO 1 | 06 |

Q. 2 Solve Any Two of the following.

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|----|--|----------|----|
| A) | Define - most economical section of channel and derive conditions for most efficient triangular channel section. | Remember | 06 |
| B) | Show that, for maximum discharge in a rectangular channel, the depth of flow is $2/3$ of the specific energy. | CO 1 | 06 |
| C) | A 2.5 m wide rectangular channel has a specific energy of 1.50 m when carrying a discharge of $6.48 \text{ m}^3/\text{s}$. Calculate the alternate depths and corresponding Froude numbers. | CO 1 | 06 |

Q. 3 Solve Any Two of the following.

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|----|--|----------|----|
| A) | State the gradually varied flow assumptions and derive the dynamic equation of gradually varied flow. | Remember | 06 |
| B) | Enlist and give description about types of hydraulic jump. | CO 2 | 06 |
| C) | A jet of water, 10 cm diameter moves with a velocity of 25 m/s and strikes a series of flat plates fixed on the periphery of wheel. If due to impact the wheel rotates at 100 rpm, calculate (i) Force exerted by jet on plate
(ii) Work done on the plate/sec
(iii) Torque exerted on wheel if the radial distance at which jet strikes the plate and axis of wheel is 1 m. | CO 2 | 06 |

Q. 4 Solve Any Two of the following.

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|--|-------------|-----------|
| A) Classify different types of turbines. Describe main components of Francis Turbine with neat diagram. | CO 3 | 06 |
| B) Write a note on selection of turbines. | CO 3 | 06 |
| C) A Pelton wheel turbine has a mean bucket speed of 9 m/s and discharge of 680 lps under a head of 35 m. The bucket deflects the jet through an angle of 165° . Calculate power and efficiency of turbine. Assume $C_v=0.97$ and blade's surface to be perfectly smooth. | CO 3 | 06 |

Q. 5 Solve Any Two of the following.

- | | | |
|---|-----------------|-----------|
| A) Explain 1) Efficiencies of a Centrifugal pump
2) Airlift pump and submersible pump | Remember | 06 |
| B) How centrifugal pumps are classified? Explain in detail working of centrifugal pump. | CO 3 | 06 |
| C) The impeller of a centrifugal pump runs at 1440 r.p.m. and has a vane curved back at 30° . The radial velocity of flow is constant at 3 m/s. If the manometric efficiency is 80% and the manometric head is 25 m, determine
(i) the vane angle at inlet
(ii) the diameters of the impeller at inlet and exit
Assume the ratio of outlet diameter of runner to inlet diameter as 2. | CO 3 | 06 |

----- **End** -----